

Metering Machaut's *Chansons baladées*

Introduction: Danced Songs and Medieval Meter

Guillaume de Machaut (1300-77) composed dozens of songs in each of the *formes fixes* of his day: 40 ballades, 20 rondeaux, and 33 virelais are set to music in Ludwig's classic 1926 edition.¹ Rather than "virelai," however, Machaut was conscious and persistent in his use of the alternative term *chansons baladées* ("danced songs"). In the poet's words: "I began this virelai / which is called a danced song; / thus it should be proclaimed" (*Remede de Fortune* lines 3448-50, trans. in Peraino 2011, 236). Machaut's insistence has suggested to modern scholars an effort to cling to the genre's dance origins.² And from that origin it is generally accepted that fourteenth-century dance song exhibits simple, consistent metrical groupings. Lawrence Earp, for example, states that "we hardly need to mention that the rhythmic projection of such a song would have been strictly metrical, something to move the feet to" (Earp 1991, 124). David Maw observes that "metrical organization is a necessity for dance music, as the coordination of bodily movements requires audible marking of the bar in the construction of

¹ See Earp 1995, 241-3, on the difficulty of enumerating Machaut's lyrical output by genre given the variety of their manuscript sources.

² Peraino 2011, 236, points out the same terminological insistence in Machaut's *Prologue*: Section V, lines 14-15. She attributes Machaut's connection of the virelai with dance to French refrain songs called *rondets de carole* and *ballettes*, and to the Occitan *dansa* (248). According to Earp 1991, 115, "it appears that Machaut was quite self-consciously preserving this genre [virelai] for the dance." He traces the development of the danced song genre in Machaut as a generic mixing of high courtly poetry with low metrical dance. Referring to Machaut's virelai "Dame, a vous sans retollir," Leach 2011, 159, notes that these songs were, for him, "very specifically a social dance form."

the music” (Maw 2004, 61). This paper continues Maw’s work in examining what that metrical organization might be, especially when the notation admits of various metrical interpretations. I build upon Maw in probing hierarchical layers of meter from surface to large (hypermetrical) spans, in construing relationships between tonal exploration and metrical instability, in consideration of how *formes fixes* repeats can upend (and sometimes reset) an established metrical environment, in interpretation of motivic development against metrical regularity, and in some speculation about how certain notated rests may be an artifact of a dance tradition. The great majority of Machaut’s “danced songs” are notated in binary mensurations, enough to establish a trait for the genre. But their tonal organization and other kinds of musical articulation enumerated below often do not rigidly or entirely conform to a fixed binary pattern throughout. The means of rhythmic notation, that is, do not necessarily cause the sense of metrical grouping in a work. In the analyses to follow, mensural division is not regarded as the ultimate arbiter of metrical grouping but as a notational background against which independent metrical play can emerge as a song unfolds.³ It is possible, in other words, for entirely regular mensural notation to be interpreted in flexibly irregular metrical groupings when prompted by divergent parameters of musical articulation.⁴ Yet it is because of an operative norm that the metrical play I discuss in these songs becomes particularly salient and ripe for interpretation.

³ Consult Table 1 of Maw 2002, 76, for a list of mensurations and cadential plans in the 88% of Machaut’s virelais that are set in imperfect *modus* and *tempus*. The small subset of triple time works, to be considered below, are not represented there.

⁴ Examples of the flexible metrical interpretation of regular mensural notation can be found throughout Maw 2004 and 2013, including transcriptions in mixed meters.

Meter, as alluded to above, is not explicitly indicated by the fourteenth-century French system of notating discrete lengths of pitch and rest. A brief review follows in order to distinguish the production of medieval mensuration from the modern reception of meter in Machaut's music.⁵ Each duration of the mensural system can be subdivided into (or is comprised of) the next lesser duration: three when perfect, two when imperfect. Such a top-down view of rhythmic subdivision, rather than an additive conglomeration of values from smallest to largest, is adopted as preliminary to the interpretation of meter undertaken here, especially the parsing of free rhythm equivalent to larger mensural values. The largest of the three common mensural levels is called *modus*, the subdivision of longs by breves. The central mensural level with which we will be concerned is called *tempus*, defined by the subdivision of breves by semibreves. In the present transcriptions, the measure is a breve's duration and the semibreve beat represented by a quarter note (dotted when perfect). *Prolatio* is the subdivision of semibreves by minims, transcribed here as eighths per (dotted) quarter. The metrical grouping of mensural rhythm is a qualitative interpretation that accrues from the regular, hierarchical grouping of pulses. Once established, a metrical grouping can be felt to remain in effect until its pattern of accentuation is contradicted by some alternative. Following Maw's foundational work on meter in Machaut's songs, I identify the sense of medieval meter

⁵ Boone 2000 undertakes a detailed investigation of regular rhythmic punctuation in the layered mensural pulses of Machaut and others across the medieval and Renaissance periods. Boone then contrasts mensuration as temporal framework with common practice meter. Succinctly put by Maw 2002, 70, "while the theory of mensuration supplied a quantitative understanding of musical time sufficient to make sense of rhythm for the purposes of notation, metrical thought provided a supplementary qualitative understanding of musical time necessary in practice to make sense of rhythm for the purposes of musical effect."

as a combination of motivic initiation and termination, rhythmic grouping (especially recurrent), melodic phrasing, formal boundaries, agogic accent, text alignment, simultaneous articulation (in polyphony), tonal accent, contrapuntal resolution, and harmonic change.⁶

As a prefatory example consider “Tres douche dame,” a refrain interpolated into Jacquemart Gielée’s romance *Renart le Nouvel* (1291-97). A modern transcription follows as **Example 1.**⁷ While thirteenth-century musical interpolations may not always represent functional dance songs,⁸ this refrain’s musical structure well embodies Johannes de Grocheio’s

⁶ Maw 2004, 62 and *passim*, addresses the *modus* barring methodologies and inconsistencies in the complete editions of Machaut’s secular songs by Ludwig and Schrade. See Maw 2002 for a systematic investigation of *modus* in Machaut’s virelais, a more flexible mensural level that requires analytical interpretation of musical groupings, supplemented by a focus on text setting and meter in Maw 2013. A complementary analytical account of flexible *modus* barring in Machaut’s motets is undertaken in Lavacek 2022.

⁷ The author’s transcription of F-Pn fr. 25566, f. 129r. In Mullally’s 2011, 119, translation: “My very sweet lively lady, listen to my heart beseeching you.”

⁸ Dillon 2015, 615, is clear that music interpolations “were not texts inviting performance or demanding connection with a specific melody, but rather were literary representations that transferred the medium of performance into the medium of writing, where sonic associations fade.” For Rimmer 1991, 64, the *carole* cues a “combination of particular kinds of spatial pattern and particular kinds of group-to-individual relationships in music and text, with particular kinds of social function.” See Butterfield 2002, 42-63, on the sourcing and function of dance song in thirteenth-century romance and 64-71 on the narrative ramifications of evoking a certain performance practice through such interpolations. The classic source Boulton 1993, 285, discusses the function of some interpolated *rondets de carole* qua music to convey the “ease and gaiety of the imperial court.” The same interpolations also reveal cultural values in the important nostalgic project of preserving a corpus of songs through literary record (275), perhaps especially for unicum refrains like this one. Huot 1987 details a shift from orality to literarity in the fourteenth century as manifested by a greater focus on the genesis of interpolated lyrics rather

c. 1300 description of the particular dance song genre considered here and Nico H. J. van den Boogaard's essential refrain catalogue (vdB 1797) lists its purpose as caroling, an association continued in the dance scholarship of Robert Mullally (Mullally 2011, 119, and 1986, 230).⁹ The melody's steady trochees (methodically embellished), its melodic pattern, and its predictable cadential plan contribute to a regular metrical grouping, indicated here through unobtrusive bar lines.



Example 1. Gielée's "Tres douche dame," a refrain interpolated in *Renart le Novel*

than on their performance, satisfying a growing demand for visually appealing books for reading, less as repositories of words to be spoken aloud. Of primary interest, then, is not performance of an interpolated song. Rather, the illuminated manuscript itself has the performative quality of staging its own events (4). For this Huot draws due attention to the medieval scribe, compiler, and illuminator as *metteur-en-scène* of all the visual elements of a book. An introduction to the medieval romance with musical interpolations is Coldwell 1981.

⁹ McGee's valuable study of Grocheio's survey of secular dance music at the turn of the fourteenth century equates the musical genre named the "vocal ductia" with the *carole*, a connection he furthers by citing poems of the same form called "carols" in fifteenth century England (McGee 1989, 506-11). Peraino 2011, 248, attributes Machaut's connection of the *virelai* with dance to French refrain songs called *rondets de carole* and *ballettes*, and to the Occitan *dansa* (248). Chaganti 2018, 196, is clear that "song in *virelai* form came to replace the *rondeau* as an accompaniment to *carole*." Machaut himself makes the connection between dance and the *virelai* in the *Remede de Fortune*, discussed below.

This love song is comprised of two parallel phrases, both essentially stepwise. The antecedent, mm. 1-3, is a series of tumbling cascades to an open (*ouvert*) cadence on E.¹⁰ The consequent, mm. 4-6, is exactly parallel, finishing a note sooner with a closed (*clos*) cadence on F. Such music was aptly described by Grocheio in *De musica*, for whom the *carole* [*ductia*] is “a light dance song, rapid in its ascent and descent, which is sung in carol [*choreis*]” and which pairs open and closed endings (McGee 1989, 506). Grocheio goes on to make a pertinent distinction between this lighter genre and another vocal dance genre, the *estampie* [*stantipes*], which he considers more difficult because it lacks the *carole*’s regular metrical stress.¹¹

The simple periodicity of this song is a basis for contrast with the musical complexity of Machaut’s *chansons baladées* a generation later. A number of modern scholars have pointed to Machaut’s metrical ingenuity in the genre. Maw finds the irregular (*modus*) metrical schemes of Machaut’s *virelais* “unsuitable for dancing except of a stylized nature” (Maw 2002, 71) while Lawrence Earp points out the “fashionable *ars nova* rhythms” of their surface (Earp 1991, 125). Judith A. Peraino similarly finds the relatively “off-kilter rhythmic environment” of Machaut’s *virelais* to result in “a lively tune that dances itself, though it may not be itself fit for dancing” (Peraino 2011, 356). And Daniel Leech-Wilkinson notes that “hardly any have the regular metrical structures one would expect of a dance accompaniment” (Leech-Wilkinson 1991, 16). While naming the genre for its dance origins, Machaut developed the *chanson baladée* through polyphonic accompaniment and, as I will show, a metrical ingenuity that likely

¹⁰ An open (*ouvert*) cadence is identified with reference to a song’s terminal pitch and implies future resolution in a closed (*clos*) cadence. Scale degrees ^2, ^5, and ^7 are customary sites of open cadence, all ultimately seeking ^1.

¹¹ McGee 1989, 509-10, and Mullally 1998, 22.

overflowed the modest cup of public round dancing.¹² Even if the *chanson baladée* transitioned from dance song to art song in Machaut's hands, we have the composer's word paired with an illustration in the earliest complete works manuscript overseen by the composer (Mach C, without in Mach E, F, and Vg) that this genre was connected with round dancing; at least he wanted it represented that way.¹³

Consider the danced song "Dame, a vous sans retollir," from Machaut's *Remede de Fortune* (c. 1340s).¹⁴ While the musical interpolations of the *Remede* include all the *formes fixes*, Machaut associated dancing with the *chanson baladée* alone by setting an illustration of

¹² Earp 1991, 113-14, credits the infiltration of rhythmic innovations applied earlier to *ars nova* motets into secular song with its dissociation from dance. Earp 2005, 106, "believe[s] that the early virelais of Machaut are better understood as highly artistic works that merely evoke characteristics of the old dance genre."

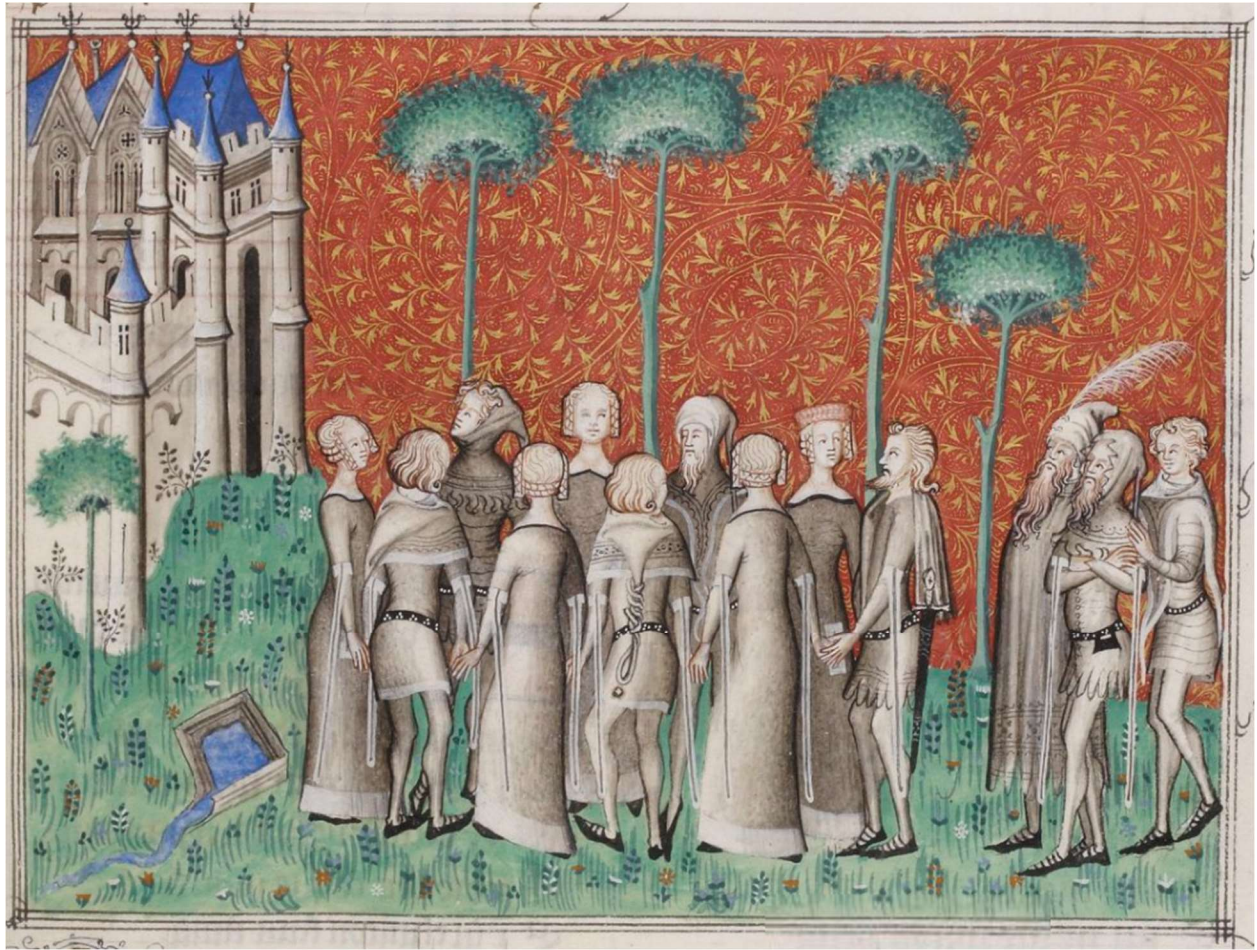
¹³ Nevile 2008a, 30, and Dickason 2021, 177, give recent summaries of the common description of caroling in the round. Morrison 2021 extrapolates the carole's circular movement to a choreographic structuring of *Sir Gawain and the Green Knight's* (14th c.) strophic form. That the *carole* could also be a line dance progressing through the streets of town is noted by Eustace and King 2011, 52-56, Page 1990, 115, and McGee 1989, 507. Dance formation has no bearing on my speculation here. Set beside the other songs of *Remede* in Mach C are depictions of people holding a scroll and sometimes a pen to illustrate reading and composing. On issues of class, gender, and social adhesion in the iconography of round dancing in Machaut's *Remede* and other medieval narratives, see Alexander 1996, esp. 151-53.

¹⁴ See Peraino 2011, 239-42, for detailed elucidation of this song's lyrics and function within the *Remede*.

danced performance beside the score, given as **Example 2**.¹⁵ Here, carolers sing and dance a virelai led by its composer.¹⁶

¹⁵ Earp 1991, 126, claims that the story line of the *Remede* was contrived to demonstrate each type of song then current, while Peraino 2011, 235, calls the *Remede* “in effect a musical *summa*.” See also Earp 2005, 105-6, on the virelai as danced song in the *Remede*. Dickason 2021, 174ff, explores the *carole* dance’s role in sacralizing courtly love in the medieval romance *Roman de la Rose*. See Butterfield 2002 for the generic characteristics of *romans à chansons* and Stokes 2014 for the didactic undertones of Machaut’s *Remede*, particularly regarding medieval rhetoric. Leach 2011, 139-64, provides an excellent summary of the *Remede* and analysis of its lyric interpolations in the context of the poetic functions of Hope and Desire. Boulton 1993 defines six different functions of lyric insertions into a variety of medieval French literary works, including Machaut’s *Remede* (188-92).

¹⁶ According to Peraino 2011, 285, “there we see the singer Machaut hand in hand with women in the dance. His voice in the *virelai* enables the carole.” See also Leach 2011, 159, on Machaut’s identity in this song and its depiction. Leo 2005, 66-68, explores iconographic relationships between Machaut manuscripts and those of *Le Roman de la Rose*, including scenes of caroling in the *Remede*. Straple-Sovers 2023, 597, cautions that we not read too much into medieval illustrations of dance, “since these sources were not created with the intent to accurately record physical movements, relationships of dancers to one another, or their movement through space.” The Machaut manuscripts are artistic objects formed in a tradition of presentation copies, in other words, and not a photograph capturing actual practice.



Example 2. dancing Machaut's virelai "Dame, a vous sans retollir;" detail from *F-Pn* fr. 1586

(Mach C), 51r

A modern transcription of the song follows as **Example 3**, including metrical beat counts at the *tempus* level and formal labels A and B.¹⁷ The A section (refrain) opens with tonal

¹⁷ My transcriptions of Machaut are based upon Ludwig 1926 unless otherwise noted, differing sometimes in suggested ficta and bar line placement. In the case of "Dame, a vous sans retollir," I prefer the anacrustic transcription of Maw 2002, 82, for its benefit of situating both cadences on downbeats, contra Ludwig and Schrade. My beats 1 and 2 in Example 3 correspond to what Earp 1991, 129, calls "metrical units" 1 and 2, adjusted for the anacrusis. Earp 2005, 106-7, uses the same song as an example of conventional text declamation. His bold-numbered syllables there correspond to my numbered beats 1 and 2 both.

articulation of G (mm. 2 and 5), which become subordinate to the final cadence down a step on F (m. 11). The B section floats around C, an open $\wedge 5$ in reference to the final.

The image shows a musical score for Machaut's "Dame, a vous sans retollir;". The score is written in 8/8 time and is divided into three sections: A, B, and C. Section A (measures 1-6) is marked with a treble clef and a key signature of one flat. It features a melody with various rhythmic values and is annotated with metric numbers (1, 2) above the notes. The lyrics are: "Da-me, a vous sans re-tol - lir | Dong cuer, pen-se-e, de- sir, | Corps, et a - mour, | Com-me a tou - te la mil- Vos - tre biau - te fait ta - rir | Toute autre et a - ni-en - tir, | Et vo dou - cour | Pas - se tout; rose en cou-". Section B (measures 7-10) is also marked with a treble clef and a key signature of one flat. It continues the melody and is annotated with metric numbers (1, 2) above the notes. The lyrics are: "lour | Qu'on puist choi - sir, | Ne qui viv - re ne mo - rir | Puist a ce jour. - lour | Vous doi te - nir, | Et vo re - gars puet ga - rir | Tou - te do - lour. |". Section C (measures 11-14) is marked with a treble clef and a key signature of one flat. It features a melody with various rhythmic values and is annotated with metric numbers (1, 2) above the notes. The lyrics are: "Si ne me doit a fo - lour | Tour-ner, se je vous a - our, | Car sans men-tir, | Bon-te pas - ses en va - lour, | Tou - te flour en douce o - dour | Qu'on puet sen-tir. |".

Example 3. Machaut's "Dame, a vous sans retollir;" metrically interpreted

Compared to the *Renart* chanson (Example 1), Machaut's dance song is longer, traverses a wider range, dabbles in chromaticism, and freely mixes trochaic and iambic subdivisions.¹⁸ In support of the anacrusic transcription here, the song's cadential layout is representative of the genre in tying cadential arrivals, text line endings, and formal closure to the same one of two semibreve locations in imperfect *tempus*. Thus, what I call beat 1—the site of metrical accent—emerges as the stronger beat in a song that does not begin with it. Metrical accent, then, is a property ascribed to the (mensural) notation of undifferentiated pulses. These accents demarcate grouping structures through the coincidence of musical and textual parameters such as those named above. The emphasis of the beat designated 1 here is a strong and

¹⁸ For Earp 1991, 138, "Machaut's early dance virelais... are already far more refined than a popular dance song."

characteristic correlation in medieval dance music. Machaut’s virelais cadence on beat 1 91% of the time out of about 120 cadences.¹⁹ My perception of beat 1 carrying the bulk of cadences may seem inevitable since I rely upon cadential placement (among other features) to mark beat 1s in the first place. The correspondence of cadential arrival with emphasis of other kinds—modal weight, agogic stress, lyric alignment, rest position—allays the danger of tautology here.

The generic linking of a particular metrical beat to tonal emphasis provides an illuminating tool for analysis. For example, consider the unusual conclusion of the B section (m. 11) in Machaut’s danced song “Loyaute weil tous jours,” given as **Example 4**.

A 1 2 1 2 1 2 1 2 1 2 1 2 1 2

Loy - au - te weil tous jours mein-te - nir | et de cuer ser - vir | ma da-me de - bon - nair - re. |
 Son tres doulz vo - loir sans re-pen-tir | Et li o - be - ir | Com a mis, sans m'ef - fai - re. |

B 1 2 1 2 1 2 1 2

Mon cuer y weil et mon de-sir | met - tre sans re - trai - re. |
 Ne ja ne m'en quier de-par-tir, | Eins weil tou-dis fai - re |

Example 4. Machaut’s “Loyaute weil tous jours” (V2); metrically interpreted

The pitch center of this song is F, ratified by the refrain’s final cadence (m. 7). The unequal line lengths of the poetry are distributed by the melody to end mainly on tonal pillars F

¹⁹ Cadence here refers to arrival on a pitch stressed by alignment with the end of a poetic line, agogic accent, being followed by rest, modal relationship to the last note of the song, and by perfect consonance with other voices in polyphonic cases. Moll 1998, 32, provides a useful table for defining cadence in fourteenth-century French masses.

and C, once touching on A (m. 5) on the descent through the Lydian pentachord. Singing three times through the refrain of this virelai establishes F (^1, the modal final) as a beat 1 kind of note. Notice, then, that the verses also end on F (m. 11), but situated on beat 2. In a typical *forme fixe*, the B section terminates on an *ouvert* scale degree, a pitch that calls for tonal closure later with the return of A.²⁰ The regular metrical grouping 1—2 / 1—2 provides a medium through which to interpret weakness in the B melody's hop down to final F.²¹ What appears melodically astylistic in “Loyaute weil tous jours”—a short verse ending on the final—is redeemed metrically because ^1 is relegated to the conventionally weak beat 2. In other words, a pitch that is melodically *clos* is positioned as metrically *ouvert*.

Metrical Organization in the Danced Songs

I now turn to Machaut's *chansons baladées* in which irregular groupings, subverted metrical entrainment, and off-kilter metrical organization represent the development of a dance genre rooted in steady metrical procession. As with the previous dance songs in imperfect major mensuration, Machaut's virelai “J'aim sans penser laidure” can be transcribed in 6/8 and metrical identity assigned to each beat like in **Example 5**.

²⁰ Some B sections in other of Machaut's virelais have paired open/closed endings in which the second ending, that which closes that B section, does end on the final. The same open/closed tonal relationship plays out in the shorter V2 across the A and B sections rather than within the B section itself.

²¹ This weak-beat F is further attenuated by the natural declamation of the poetry's feminine ending, where the penultimate syllable of “re-TRAI-re” is stressed (beat 1) and the final syllable unstressed (beat 2).

A

8 J'aim sans pen - ser lai - du - re | et ay lonc - temps a - me | celle ou dieux et na-

8 tu - re | ont mis tant de bon- te | que tou-te cre-a - tu - re | d'on-neur a seur-mon - te |

Example 5. Machaut's V14, "J'aim sans penser laidure" (refrain only); metrically interpreted

The recurrent trochees and iambs, regular agogic accents (mm. 1-6 especially), and tonal articulations (mm. 5 and 14) contribute to a regular binary metrical grouping in this song. Relatively long strings of eighth notes, such as in m. 8, however, having no dependence upon a neighboring quarter to capture them as trochees or iambs, can be susceptible to alternative metrical groupings if pitch and lyric articulation reorganize them. While a change in mensuration is not represented in Machaut's notation, the grouping of these rhythms—the question of meter—is open to interpretation. The opposing groupings 2x3 and 3x2 being weighed here are of equivalent duration, a demand medieval mensural theorists called equipollence. This single breve of disruption represents a bump in the road, not a change of path. Example 5 follows the metrically flexible edition of Sylvette Leguy in rendering a text setting of one syllable on each of three equal beats in m. 8 and again in m. 12, the same motive with textual assonance.²² Leguy's 1977 complete works edition (from Mach G) is remarkable

²² Following Earp 1991, 130, "there are no extended melismas, and thus the ear hears the text syllables themselves as providing the musical rhythms." The syncopated "pitch accents" marked in Example 1 of Maw 2002, 72, occasionally insinuate a perfect minor grouping that rubs against the prevailing imperfect major metrical and declamatory accents of Machaut's V1, "*He! Dame de vaillance*." Example 14c of Maw 2002, 91, explicitly renotates

indeed for its flexible approach to metric grouping as a response to Ludwig's supposed rigidity.²³ Whereas Ludwig and Schrade beam mm. 8 and 12 as continued iambs in 6/8, the present study seeks to explore the ramifications of flexible metrical grouping in the *virelai* genre, a trend gaining modern currency. David Maw's recent work (2013, 2004, 2002) explores a similar responsiveness to irregular metrical/textual groupings in Machaut's songs, and his transcriptions of some *chansons baladées* also feature rebarrings and changes of time signature such as I have ventured here.

In what follows, I will consider evidence for metrical regrouping in the refrain of V14. The tight spacing of the consecutive eighth notes in mm. 8 and 12 followed by a generous gap before the final quarter note—demarcating beat 3 of 3/4 rather than breaking up an iamb of 6/8—in all manuscript sources can be suggestive of the grouping change I am pursuing. But this spacing, the system breaks (Mach C), and even the page breaks (Mach A) that occur in some manuscripts between the last F and A of mm. 8 and 12 could simply reflect the line break in the poem (indicated by vertical stroke in the examples), inked in before the music notation was added. However, the way that Machaut repeats the structural pitch A in mm. 7-8, as well as

V18, "*Helas! et comment,*" along the same lines, switching to 3/4 to capture a measure of rhythm identical to that of mm. 8 and 12 of V14 as I have rendered them in Example 5. Example 8 of Maw 2013, 397, presents essentially the same transcription of V18 in 6/8 (mm. 1-6) then 3/4 (mm. 7-11) based upon the syllabification of those phrases. Peraino 2011, 242, mentions a similar "metrical ambiguity (toggling between 6/8 and 3/4)" in Machaut's "*Dame, a vous sans retollir,*" examined above, without further specification.

²³ Along with V14, Leguy 1977 assigns the dual time signature 6/8 and 3/4 to *virelais* 2, 5, 8, 16, 19, 26/23, 27/24, 31/28, and 32/29 (numbered as in Ludwig/Schrade), among others with multiple time signatures, and freely beams groups of two and three eighth notes in order to capture different "rhythmic cells" (see Foreword, VI).

details of text declamation, *can* be used to define logical metric groupings.²⁴ In V14's melody, the final D (^1) is decorated with lower and upper neighbors before descending through the Dorian tetrachord D-A, closing on ^5 in an *ouvert* cadence (m. 5). In m. 7, A marks the onset of each of two beats (in 6/8), while in m. 8 the same A can mark each of three beats (a change to 3/4); the same duration segmented differently. The word "*na-TU-re*" ends the third line of the poem. In Machaut's rhythmic setting, "*na-*" is set as a pickup note to the stressed "*-TU-*" on the metrically accented beat 1 (m. 8). The weak syllable "*-re*" follows naturally on a weak beat—beat 2 in both Leguy's 3/4 reading and in Ludwig and Schrade's 6/8 reading, but aligned one eighth note later (as suggested by the manuscript spacing)—and this could have ended the measure. Yet Machaut sets another weak syllable, "*ont,*" in m. 8 (beginning the fourth line of the poem) in a way that acts as a pickup into m. 9, suggesting a third beat in the same measure, beginning a descent through the Dorian pentachord A-D.²⁵ Mm. 11-12 are slightly different. F marks the two beats of m. 11 before A returns to mark the three beats of m. 12, where again the final A is detached as a pickup to the final Dorian descent A-D.

I use an open bracket above each of the anomalous triple counts to signify a change from an entrained meter. My analytical use of the open bracket is adapted from the modern transcription of medieval coloration, in which a change of ink color or style of notehead in a

²⁴ Earp 2005, and its 2012 extension, document the conventional relationship between metrical pulses and rate of text declamation in Machaut, regarding irregularity ("declamatory dissonance") as a strategy of poetic expression.

²⁵ This Dorian pentachordal descent is balanced either by the Dorian tetrachordal descent D-A described above (mm. 1-5) or by the pentachordal descent E-A defined by the poetic line breaks, in the latter case linking the two locations of Dorian pentachord.

source manuscript indicates a change of mensural reckoning.²⁶ The switch to triple grouping occurs again in m. 12, bookending a passage between them (in gray) marked by tonal tension. This passage is the first time the song's final D is broached in its proper register (m. 9). The song has steadily descended the Dorian octave to this point, yet this low D, if the modal final, is not the refrain's conclusion. Descent continues to a low C in the following measure. Set as a pickup note at the end of m. 10, low C portends resolution to a more conclusive final D in a Dorian context. Following the second triple group at m. 12, which I hear as metrically restorative rather than destabilizing a second time, the same descending pentachord to final D is tried (m. 12) and successful closure attained in m. 14, the end of the refrain. The drama of metrical disturbance in these songs resonates with Machaut's courtly love poetry in the *chansons baladées* genre. The lyric voice of Machaut's virelais often professes being at great pains to maintain steadfast loyalty to a beloved paragon, whatever may come. In setting his lyric poetry to music, Machaut allows for higher level meanings to emerge as one medium reflects upon the other. Reevaluating the genre through a lens of metric flexibility, I propose that grouping disruptions can suggest departure from or a trial of the beloved's professed steadfastness, which, in the case of V14, is ultimately overcome and metric normativity restored together with tonal closure.

Equipollent metrical swaps like that of V14 can also be found in Virelai 27/24 (Ludwig/Schrade), "En ma cuer," only reversed. In this song, the prevailing meter is triple and

²⁶ Open brackets in modern transcription are to be distinguished from closed brackets, which usually represent ligated neumes, a means to convey rhythmic values of a set of pitches.

the disruptions duple.²⁷ Besides its triple meter, this virelai is also atypical in being polyphonic.

The inclusion of counterpoint likely represents further evolution beyond the genre's practical dance origins.

The musical score for Machaut's V27/24, "En ma cuer" (refrain only), is presented in two systems. The first system contains measures 1 through 12, and the second system contains measures 13 through 24. The score is written for a cantus part (top staff) and a tenor part (bottom staff). The cantus part features a complex rhythmic pattern with many triplets and some duple measures. The tenor part provides a harmonic accompaniment. The lyrics are: "En ma cuer a un des - cort | qui si fort le point et mort | que, sans men - tir, s'A-mours par son | doulz plai - sir | n'i met ac - cort | a-veuc ma da - me, pour mort | me doy te - nir." Above the cantus staff, there are two rows of numbers indicating the metric interpretation of the rhythm: the first row shows groups of three (1 2 3) and pairs of two (1 2), while the second row shows groups of three (1 2 3) and pairs of two (1 2). Some of these numbers are in gray, indicating metric uncertainty.

Example 6. Machaut's V27/24, "En ma cuer" (refrain only); two options metrically interpreted

In **Example 6**, striking pairs of iambs on the surface are notated in 6/8 (imperfect major) on a separate staff above the score set in 3/4 (perfect minor). In the iambs' first instance (m. 3), the accompanying tenor reinforces the cantus' lilt for a single measure. An iambic reading yields only two beats per measure and I again interpret the interruption of an entrained pattern as heralding a passage of metric uncertainty (in gray above, compare to 1-2-3 in black below). Just as in V14, metric uncertainty is coordinated with tonal tension. The modal final is broached for the first time within this metrically uncertain passage (as in V14), now F in m. 4. But the finality of F is quickly undercut by the accompanying tenor's desertion of the perfect

²⁷ Fuller consideration of those rare virelais set in triple time is taken up below.

octave F/f to skip up to A, forming an imperfect sixth that pulls weakly towards G. The cantus then takes up a new phrase a tritone away, abandoning closure on F to begin again on B (m. 5). Another equipollent grouping change in m. 7 (in the cantus alone) restores a sense of metric normalcy, defining again a parenthetical passage (mm. 3-7) in which tonal closure is tried but not yet fully realized. Immediately upon what I am calling the restoration of metrical normalcy, the melody cadences definitively on octave Cs (^5), the Lydian mode's tenor.

Another metrical swap in m. 9 denies comfortable entrainment in the overarching grouping pattern, marking off a passage (mm. 9-13) that again concludes with a cadence to an *ouvert* scale degree. Arrival this time is on a tonicized G (^2) in m. 14, another common place for medial cadences in the Lydian mode, echoed quite strongly in m. 16. The regularity of preparing cadential articulation with equipollent metrical play by now suggests a formal device in this song at least. The same motivated A/F(♯) dyad appears at the beginning of a new phrase in m. 17, but the G it targets is a faint reminiscence as ^2 is subordinated to the final F that closes the refrain (m. 22). The descending Lydian pentachord C-F in the cantus aligns with the ascending Lydian tetrachord C-F in the tenor (mm. 19-22) to affirm the hegemony of F, hinted at only once before in the uncertain pocket of m. 4.²⁸

Metrical ambiguity looms, however. This final F (m. 22) is approached in the cantus through a measure of iambs identical to m. 3 (see m. 20), interpreted above as destabilizing.

²⁸ These are highlights of a larger tonal arch fashioned by the poetry's line lengths. Set in groups of 7 and 4 syllables, as marked with vertical strokes in the example, the lyric endings define an orderly plan initially rising in fifths F-C-G (mm. 4, 8, 10), pausing on G (m. 14) as an open neighbor to final F, then descending back through fifths G-C-F to close (mm. 16, 20, 22).

There are five such measures of ambiguous iambs in the cantus and if they are read consistently across the virelai's *formes fixes* repeats, the song seems to end on uncertainty. The refrain is repeated three times in the standard fixed form AbbaA, in which letters A and B represent different music (refrain and verse) and lower case indicates new lyrics set to the same music. When the refrain is set to new lyrics (lower case a), the passages of uncertainty just discussed will swap places with what was stable the first time through. And then, fittingly, the refrain will revert to normal upon the third and final statement (capital A), when the original lyrics are restored to the refrain and the piece ended. But that means, of course, that the song ends with uncertainty. Rather than being a dissatisfying conclusion, such a reading pairs honestly with the protagonist's closing sentiment: if the *dame* remains unmoved by Love, it will be the death of him (*s'Amours par son doulz plaisir n'l met accort avec ma dame, pour mort me doy tenir*).

The examples so far have been analyzed at the *tempus* level of mensuration, counting each semibreve, most often two beats per measure of 6/8. In my analysis of V20, I will group measures (of 6/8) at the broader *modus* level of mensuration, akin to hypermeter in later European music, and retain the strong-weak relationship of beats 1-2. The parsing of *modus*, a measure of breves per long, requires an act of analytical interpretation when longs are not present in a melody's notation.²⁹ This study continues modern analytical work on medieval meter by considering mensural levels to be expressed not only by discrete notated rhythms but by free rhythm that adds up to a given mensural unit. For example, a song may start with two semibreves that can be grouped as one (imperfect) breve's duration; a beat at the *modus* level,

²⁹ See Maw 2004, 60, in response to Hoppin 1960.

a measure in transcription. So too can one semibreve and three minims be grouped as one *modus* beat, the three minims together equaling a second semibreve of that measure. The simple math of these examples was given without respect to pitch or text setting. When grouping free rhythm into larger durations, the analyst is often guided by melodic sense units that express consonant intervals through neighbor or passing motions, by text alignment, by cadential pull and resolution, and, in polyphony, by blocks of harmonic identity. In **Example 7**, broken bar lines show breves (with beats counting *tempus* divisions) while full bar lines show longs (with measures counting *modus* divisions). The same kinds of grouping complications found at the *tempus* level above can be found at the *modus* and even the *maximodus* level above that, too. Interpretive preferences like locating cadential arrivals on strong beats—secondly, on any beat rather than off—and situating stressed syllables on metrically strong positions while relegating final mute vowels to weak ones can clarify broader rhythmic grouping in medieval French dance songs.³⁰

Those *chansons baladées* that do not feature grouping play can still be shaped by metrical organization. The form of Machaut's V20, "Se d'amer me repentoie," is enlivened by a metrical narrative that plays out across its repeats, as annotated in **Example 7**.

³⁰ Maw 2013 provides a thorough analysis of text setting in the Machaut sources, with reference to their mensural and metrical implications, and reconstructs some principles for fitting words to music in his songs.

modus: 1. 1 2 2 1 2 2 1 2 2 1 2 1

2. 2 1 2 1 2 1 2 1 2 1 2 1

A

8 Se d'a - mer me re-pen - toi - e | ne fein-gnoi-e, | trop fe - roi - e con-tre mi; |
car tout mon bien si per - droi - e | que n'a - roi - e, | ja mais

12 2. 1 2 1 2 1 2 1 2 1 2 1

B 3. 1 2 1 2 1 2 1 2 1 2 1 2

bon jour ne de - mi. | Si weil a - mer mon a - mi | loy - au - ment, ou que je soi - e,
et a - voir le cuer jo - li, | gay, chan-tant et plein de joi - e; |

Example 7. Machaut's V20, "Se d'amer me repentoie;" metrically interpreted at the *modus* level

Machaut sets his poem in units of two or four bars, for the most part, neatly expressing imperfect *modus*. But the regular pace of declamation established in the refrain, two syllables per bar, is disrupted by the first ending's protracted two bars per syllable. The melody opens with an octave's cascade from high G to low, whereupon an inflected F# overshoots the final G to land on an open A (^2) on a strong beat 1 in m. 11. If this F# leading tone resolved to G in m. 10, rather than pass through it to A in the next measure, the refrain would take the balanced shape of 4+2+4 bars. As a result of the first ending's melodic elongation to five bars (4+2+5), the strong-weak relationship of the refrain's metrical organization is exactly flipped when repeated according to the *forme fixe*. The refrain's second ending, however, is shorter than the first by a measure and, balance restored, the closed cadence on final G (m. 14) also falls on a strong beat 1. If both refrain endings were of the same length, the final cadence would land on a weak beat.

The B section sets forth on a weak beat (m. 15) opening again on high G, recalling the refrain, but only winding down to an open cadence on D (^5) in m. 23. Just as in the refrain, the beat strengths are again reversed when B repeats. This is due to a musical setting of 4+5,

resembling the A section's 4+2+5 but with the "extra" bar being at the front of the group of five. There is an incipient melodic return of the refrain's opening four bars in mm. 20-23, the extra bar being the anacrustic m. 19. The way its C# leading tone overshoots D recalls how F# overshoot G a fifth lower at the end of the A section. It seems fitting that only in the final return of the refrain, where a (same music, new text) is succeeded by A (same music, same text), should V20 return the same music and same text to their original metrical disposition. The threads of the song thus tied together, it finishes.³¹

In addition to equipollent changes of metric grouping and off-kilter formal repeats, Machaut's polyphonic virelai "De tout sui si confortee" explores metrical novelty in connection with phrase organization. **Example 8** interprets the song's refrain. Beats are labeled above the score at the *modus* level. This initial example contains only the cantus part, to discuss its structure in isolation.³²

³¹ My transcription of V20 conforms with all but one Machaut manuscript in which this song is represented (Mach B, C, E, G, and Vg). It differs only from Mach A, which appends a breve (half) rest after the *ouvert* cadence on A in m. 11. If this rest is a measured duration and not merely a section marker, it represents a significant metrical change insofar as it provides a new measure for another *modus* beat. This extra measure would realign the repeat of the refrain on the same metrical footing—that is, m. 1 would return on beat 1 as before—correcting what I characterized as a mismatch. However, I take this phantom breve to be a section marker because it is not present when the refrain is renoted with new text later in Mach A (after "oubli;" not shown in the example).

³² My transcription here is closer to Schrade (numbered V32), which differs from Ludwig (numbered V38) in maintaining duple time throughout, omitting his frequent alternations to quadruple time (notated 4/4). I also maintain duple time through breves 22-24, where Ludwig compresses the same notes into a single measure of 3/2.

2 1 2 1 2 1 2 1 2 1 2! 1

α β

8 De tout sui si con-for-te-e | que ja mais n'iert ho-ste-le-e | tri-stes-se n'e-smay |

2 1 2 1 2 1 2 1 2 1 2

α β

13 8 en mon cuer, ain-cois a-ray | lie et jo-li-e pen-se - e, |

1 2 1 2! 1

β

24 8 tant com je vi-vray. |

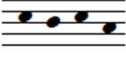
Example 8. Machaut's V38/32, "De tout sui si confortee" (cantus line of the refrain only);

metrically interpreted at the *modus* level, phrase organization labeled

The refrain of "De tout sui si confortee" is essentially comprised of two phrases—labeled α and β in the Example to distinguish them from formal labels A and B—varied in their repetitions melodically, rhythmically, and metrically. Simple beat counting foregrounds Machaut's elastic approach to repetition here, as the same melodic skeleton is realized across longer and shorter time spans. Phrase α circulates upper and lower neighbors around the pitch C, ultimately cadencing on that same open $\wedge 5$ (mm. 4-5). Phrase α is prepared by what I take to be a pickup (m. 1) before the first downbeat. Locating the downbeat on m. 2 best responds to the song's musical structure for a few reasons. First, doing so ensures that the melodic, textual,

I will later achieve the same effect, with the benefit of placing the final F on a downbeat (breve 27), by interpreting perfections at the *modus* level, just as Ludwig implies by his 3/2 time signature within the context of 2/4 and 4/4.

and agogic accent on the cadential C in m. 4 is corroborated with the strong beat. Second, a very close repeat of phrase α occurs without the pickup in mm. 14-16. Third, when the refrain repeats the pickup falls appropriately on weak beat 2, preserving m. 2 as a strong downbeat.

Fourth, the transposable motive  is a recurrent one throughout Machaut's songs.³³ In this danced song it is used as an anacrusic windup targeting the C of m. 2, a modally structural pitch (the tenor) made sweeter by the prior accented passing tone B.³⁴

Phrase β consists of an essentially stepwise ascent from the song's final, F, to C and back to down again. It is natural that the tonally open phrase α would be followed by the closure of phrase β . While that is ultimately what happens, the detail of V38/32's melodic shape is a bit more complex. A summary sketch is presented as **Example 9** below, with phrases marked and cadences labeled O for open (*ouvert*), C for closed (*clos*). The song's Lydian final is emphasized by a stem.

³³ From the virelais alone see examples of this motive in V5 (mm. 4-5), V6 (mm. 15-16), V10 (mm. 14-15), V11 (mm. 4-5, 10-11, and 14-15), V12 (mm. 8-11 and 18-21, second endings), V15 (m. 18), V17 (mm. 8-9), V18 (mm. 9-10), V23/21 (m. 9), V26/23 (mm. 12-14), V29/26 (mm. 4-5), V31/28 (mm. 14-16, 25-26, and 31-32), and V37/31 (mm. 10-11 and 19-20), not to mention the numberless variants of its pattern.

³⁴ While emphasis of the pitch C enjoys a modal rationale here, the modern analyst of medieval music must always be aware of creating the tonal emphases they wish to find through the creative acts of transcription, such as the metrical fitting of bar lines and deploying of *musica ficta* as leading tones. Leach 2000, 45-46, reviews these issues in reference to fourteenth-century song. As speculated upon later in this study, dance provides another parameter of medieval performance that may regulate, and at the same time may tempt, the modern analyst into placing this or that note on or off the beat. *Caveat saltator*.



Example 9. tonal summary of V38/32 (cantus of the refrain only); phrasing and cadential weight marked

In its first occurrence, phrase β 's F-C-F scale is followed by an *ouvert* termination on $\wedge^2$ (mm. 11-12), postponing tonal closure for a later phrase. The return of phrase α (mm. 14-16) restarts the same melodic trajectory and is again followed by phrase β , to which is now appended a different *ouvert* ending ($\wedge^7$ in mm. 22-23). For the refrain's final phrase, β is stated alone and only now rests conclusively on its final F (mm. 27-28).³⁵

Recall in **Example 8** that the initial measure was taken as a pickup, the dual benefit of this metrical interpretation being the situation of the first cadence on a downbeat (m. 4) and its final mute vowel on the upbeat thereafter (m. 5). If this is fitting for the first phrase of V38/32, the remainder of the song is more ambiguous. The other four cadences of the refrain are evenly split as to landing on the strong or weak beat, with the final cadence—the one that

³⁵ The tonal closure achieved only upon repeat of phrase β (mm. 24-8) could have been achieved the first time (m. 22) by simply landing on F ($\wedge^1$) instead of on E ($\wedge^7$). The necessity of the repeat, in that case, may be attributable to text painting. Throughout the poem Machaut's protagonist scorns despair and, in mm. 18-22, dwells willfully upon thinking only happy thoughts across an extraordinary melisma of four measures. If the rate of text declamation here paralleled the first phrase β instead (mm. 6-12), the appendage of a third phrase β would be superfluous.

ultimately closes the song—ending against convention on a weak beat (m. 27). Whichever way one regards the opening measure, the melodic repetitions that represent the heart of the song are metrically inconsistent.

Probing broader metrical levels, beyond even the long of mensural notation, continues to reveal tonal congruences and irregularities in Machaut's danced songs. Just as beat 1 accrues a sense of *thesis* and 2 of *arsis* as a song unfolds, every other downbeat (or every third downbeat, in perfect mensuration) can feel weightier still and parse the pitch structure into ever larger units. As before, tonal articulation and formal boundaries point to larger metrical divisions, defining downbeats more salient than those that occur merely out of alternation with upbeats. **Example 10** is a foray into representing these deeper kinds of downbeats in Machaut, which in mensural terms reflects the difference between *modus* (breves per long, all bar lines in the example) and *maximodus* (longs per maxima, only the unbroken bar lines). The layered interpretation of strong and stronger beats, as well as weak and weaker ones, is analogous to the layered dots in metrical hierarchy analyses of Lerdahl and Jackendoff³⁶ and the *pulse frameworks* of Graeme M. Boone more recently, and one can easily imagine smaller beats 1

³⁶ Lerdahl and Jackendoff 1983, 19, quickly associate undifferentiated beats with strong/weak metric accent, in which beats 1 and 3 of 4/4 meter are strong while 2 and 4 are weak. In the transcription of Machaut's virelais here, this corresponds to two measures of imperfect *tempus* counted 1 2 / 1 2. Counting mensural units at the *modus* level reveals the first beat 1 to be stronger than the second—parallel to beat 1's comparative strength over beat 3—because it is also strong at the *modus* level while the second *tempus* beat 1 falls under a weak beat in *modus* reckoning.

and 2 on each beat at the *tempus* level below what is shown here.³⁷ The expectation of strong-weak alternation at the deepest layers of metrical organization in Machaut's danced songs is extrapolated from norms of composition at local levels. For example, Machaut is much more likely to provide intervallic consonance on the beat at the *prolatio* level (eighths per quarter). In V38/32's substantial refrain, for example, eighth note positions 1 and 3 set consonant intervals 86% of the time (42/49, omitting rests), enough to regard strong-beat dissonances as decorative of some underlying consonance. At the *tempus* level (each measure of Example 8), every downbeat is consonant. The more revealing distinction at this level is whether the consonances are perfect (66%) or imperfect (33%), pursued below. This generic connection between strong beat and consonance provides a backdrop against which to interpret the relationships between metrical organization, tonal articulation, and formal boundaries. As will be shown, it is especially at the deeper levels of Machaut's music where metrical irregularity is most prominent.³⁸

A starting assumption of imperfect subdivision, normative of the genre at lower levels, does not meaningfully parse the tonal content of V38/32 at the *maximodus* level. Two

³⁷ See their introduction to metrical hierarchy in Lerdahl and Jackendoff 1983, 18-30. Boone 2000, 4, presents a similar "grid of pulse classes and pulse levels" in a medieval repertory.

³⁸ On flexible metrical grouping as valued characteristic of deep mensural levels in Machaut, see Boone 2000, Maw 2004, and Lavacek 2022. The flexible metrical interpretations of *modus* and *maximodus* here resonate with Lerdahl and Jackendoff 1983, 22: "this large level of metrical analysis [at 4-bars] is open to interpretation, whereas smaller levels are not."

cadences, boxed in **Example 10**, including the final one, fall between beats at this summary analytical level.³⁹

maximodus: 1 2 1 2 1 2 1 2
modus: 1 2 1 2 1 2 1 2

8 De tout sui si con-for - te - e | que ja mais n'iert ho-ste - le - e | tri - stes - se n'e - smay |

13 2 1 2 1 2 1 2 1 2 1 2
8 en mon cuer, ain - cois a - ray | lie et jo-li - e pen - se - e, |

24 2 1 2 1 2
8 tant com je vi - vray. |

Example 10. V38/32, metrically interpreted at the *modus* and *maximodus* levels (cantus of the refrain only); cadence points circled

In order to align tonal accent more closely with metric pulses, the following reading of the same virelai interprets occasional mensural expansions (*perfections*) at the *maximodus* level such that all five cadences fall on a beat, still more often a weak beat but at least not in between beats. My occasional interpretation of a triple grouping is marked for special interpretation against the normative backdrop of duple mensuration in Machaut's virelais. Mensural expansion here is simply a group of three thoughtfully inserted into the regular duple

³⁹ Corroborating my observation about the *modus* level above, the first measure really ought to be a pickup at the weightier *maximodus* level also or else only two of five cadences fall on a beat at all and both of them are weak (not marked in the example).

entrainment. While such an effect can occur anywhere, it is most stylistic and convincing either just before a cadence (as in mm. 8-10, 13-15, and 24-26 of **Example 11**, effecting a written-out ritard) or in capturing a pickup measure before a prominent downbeat (as in mm. 27-29).⁴⁰ In this way, **Example 11** offers a more flexible metric analysis for comparison with those regular beats that did not well match the phrase structure and tonal articulations of **Example 10** above. Closely allied to cadences, I note that the two oxytonic rhymes on *-ay* (mm. 11 and 27) are brought into alignment with strong metrical positions. The final pitch of the paroxytonic *pen-SE-e*, an ouvert E pointing towards the final F to come, also falls on a downbeat (m. 22), the longest melisma in the song bestows ample stress upon its penultimate syllable (mm. 19-21).

⁴⁰ Other modern scholars have pointed to how the broadening of mensuration can achieve an effect like a ritard before cadence. Adjusting *modus* and larger mensural groupings to align harmonic units and cadences is central to the analysis of Machaut's motets in Lavacek 2022. Kohn 1981, 41–2, comments upon a cadential retardation effected through mixed meters in the Dunstable motet Sancta Maria. Desmond 2018, 217 (Example 6.5), shows a change in all voices from imperfect to perfect *modus*, which acts like a cadential retardation, at the conclusion of each *talea* of the Fauvel motet "Garrit/In nova." Maw 2004, esp. 64–5, locates *modus* perfections in an imperfect context leading up to cadences in Machaut's ballade "Gais et jolis" (B35), among other examples. He later (86) calls this effect "a reverse hemiola at the cadence." Further examples from Machaut's virelais are in Maw 2002, 84, Example 9b.

maximodus: 1 ② 1 2 — 2 — 3 ①

α β

8 De tout sui si con - for - te - e | que ja mais n'iert ho - ste - le - e tri - stes - se n'e - smay |

2 — 2 — 3 ① — 2 — 3 2 — 2 — 3 ①

13 en mon cuer, ain - cois a - ray | lie et jo - li - e pen - se - e, |

2 — 2 — 3 ① — 2 (— 3)

24 tant com je vi - vray. | (repeat of A)

Example 11. V38/32, flexible metrical interpretation at the *maximodus* level (cantus of the refrain only); cadence points circled

I have interpreted metrical expansions six times in this refrain. The first occasion is in mm. 8-10, where beat 2 is extended for three counts (notated **2—2—3**) before a downbeat marks the cadence on G in m. 11, a strong point of articulation that fell between beats in **Example 10**. This perfection has the effect of phrase broadening before a significant arrival, opposite of the hurried little hemiolas on a much smaller scale in V14 (recall mm. 8 and 12 of **Example 5**; three beats in the place of two). The next metrical perfection is interpreted across mm. 13-15 for the same end; to unite the beat 1 thesis with a melodic arrival point, the cadence on C in m. 16. Measures 19-21 and 24-26 similarly prepare their cadences, the first an open ending on E ($\wedge^7$) and the last closing the refrain on the song's final, F, both now tied to downbeats. The perfection interpreted in mm. 16-18 is an expedient that sets up the usual cadential broadening in mm. 19-21, which preferably begins on a weak beat so as to land the

cadence on the strong one. The guiding intent for all these interpretive metrical adjustments is for metrical organization to correspond with tonal more agreeably, providing consistency in cadential approach and arrival found wanting the transcription's bar lines are accepted rigidly (**Example 10**).

The final measures of V38/32's refrain would not wrap around evenly on repeat—assuming the first measure is an anacrusis—without implying another *modus* perfection (mm. 27-28-1). This final expansion is shown in parentheses running off the end into a repeat of the refrain that occurs at the end of the virelai form. In fairness, V38/32 does wrap around neatly if simply counting through this song without regard for cadential placement. The methodology here, however, advocates for a metric parsing more collaborative with tonal articulation.

The cantus may carry the melody here, but it is not the complete song. Restoring the tenor allows an evaluation of harmony and counterpoint in relation to metrical grouping and tonal articulation. The polyphonic analysis of **Example 12** pursues the flexible reading above as more responsive to V38/32's musical organization.

maximodus: 1 2 1 2 — 2 — 3 1
modus: 1 2 1 2 1 2 1 2 1 2! 1

8 De tout sui si con - for - te - e | que ja mais n'iert ho - ste - le - e tri - stes - se n'e - smay |

8 TENOR

8 5 3 5 6 8 6 8 5 3 1 3 3 5 8 3 3 1 3

13 en mon cuer, ain - cois a - ray | lie et jo - li e pen - se - e, |

8 3 3 1 6 8 6 8 5 4 1 3 3 5 8 3 3 3 3

24 tant com je vi - vray. |

8 5 8 5 3 1

Example 12. Machaut's V38/32 (refrain) with metric interpretation at two levels, phrase organization, and discant reduction

The note-against-note discant reduction at the bottom of **Example 12** reveals relationships between pitch and metrical grouping.⁴¹ For example, Machaut is consistent in employing imperfect consonances on (*modus*) downbeats only in the approach of cadences, where their strong resolution tendency joins contrapuntal propulsion to the metrical broadening (*maximodus* perfections) I have suggested at mm. 8-10, 13-15, and 19-21.⁴² Machaut's counterpoint against recurring phrases α and β —recalling that a free tenor voice is composed against the fundamental cantus in secular song, the reverse of sacred cantus firmus style—also shows a high degree of consistency. In both occurrences, phrase α begins with a stable fifth expanding into an unstable sixth, which drives out by stepwise contrary motion to the nearest stable interval, the perfect octave. This 6-8 directed progression occurs twice in descending sequence in each statement of phrase α , while dissonant passing sevenths provide even greater drive for resolution on C (mm. 4 and 16). The first two occurrences of phrase β are also very similar, terminating each time on the same open dyad E/G that yearns for resolution on F (mm. 11 and 22). The final iteration of β (mm. 24-7) appears in compressed form and is harmonized differently. Replete with perfect consonance, a judicious third is

⁴¹ My discant reduction was largely produced by the simple omission of dissonances against the tenor, especially when such notes occur off the beat in passing or neighboring melodic motion. Leach 2000 provides an instructive example of modern reductive analysis grounded in medieval theory.

⁴² On the crucial concept of directed progressions in fourteenth-century polyphony, the classic source is Fuller 1992. Essentially, imperfect consonances are felt to drive towards the greater stability of perfect ones through stepwise contrary motion, especially $M3 \rightarrow P5$, $M6 \rightarrow P8$, and $m3 \rightarrow P1$. As a foundation for Machaut's *ars nova* practice, polyphonic simultaneity is very tightly linked to consonance/dissonance placement in the *ars antiqua* mensural theories of John Garland, Franco of Cologne, and Anonymous IV, as detailed in Fuller 1981.

employed only at the final cadence (3-1). Note the special contrapuntal strength of this final arrival, converging in contrary motion down a catalog of the intervallic consonances: 8-5-3-1 (mm. 25-28). This more solid setting of phrase β perhaps represents the singer's resolve to remain loyal "as long as I live" [*tant com je vivray*]. For all its polyphonic and metrically subtlety, a song like Machaut's Virelai 38/32 stands out as an art song evolved from a dance tradition it no longer exemplifies.

Danced Songs in Triple Time

Among Machaut's *chansons baladées*, only four (12%) are set in perfect *tempus*: V26/23, 27/24, 31/28, and 32/29.⁴³ The foregoing analysis made special interpretation of *modus* perfections within a norm of imperfect, and the converse principal will be carried over to those danced songs in triple time. While perfect groupings at the *tempus* level will go without remark, considerable flexibility will be called for at the *modus* level and beyond.

Example 13 presents the opening phrase (to first cadence) of two of Machaut's virelais in triple time with metric interpretation at two levels, *tempus* and *modus*. The cadential layout of these songs—aided by the opening rest and ascending iambs—strongly prompt an anacrusic metrical interpretation at the *modus* level, as shown in the example.⁴⁴ Begun with a runup to what I consider the structural downbeat in m. 2, all directed progressions in V27/24 and

⁴³ Insofar as Mullally 1998, 19, insists that "social dance of the period requires music in duple time," these few songs in triple time may be further evidence of Machaut's artistic extension of a genre with ancestry in dance.

⁴⁴ Table 7 in Maw 2004, 94-5, provides a detailed list of *modus* organization in Machaut's *formes fixes*, including flexible readings of perfect/imperfect alternation.

V32/29, except a single interior and non-cadential one (to F in m. 21), fall conventionally on a downbeat.

The image displays two musical staves, each representing a different virelais by Machaut. The first staff, labeled V27/24, corresponds to the text "En mon cuer a un des - cort". It is written in 3/4 time and includes numerical notation above the notes: 2 (1 2 3), 1 (1 2 3), 2 (1 2 3), and 1 (1 2 3). The second staff, labeled V32/29, corresponds to the text "Da-me, mon cuer em - por - tes,". It also features numerical notation: 2 (1 2 3), 1 (1 2 3), 2 (1 2 3), and 1 (1 2 3). Both staves show the cantus line with lyrics underneath.

Example 13. beginning of Machaut’s virelais “Dame, mon cuer emportes” (V32/29) and “En mon cuer” (V27/24); cantus only, metrically interpreted at *modus* and *tempus* levels

An annotated score of the cantus line of V27/24, omitting tenor accompaniment, follows as **Example 14**. The metrical level of analysis is now *modus*, compared to **Example 6**’s *tempus* reading of the same song. The beats of directed progression arrivals are circled.

A

En ma cuer a un des - cort | qui si fort le point et mort | que, sans men-

- tir, | s'A-mours par son doulz plai - sir | n'i met ac - cort | a-veuc ma da-

me, pour mort | me doy te - nir. | C'est de mon loy - al De - sir | qui me

wet fai - re ge - hir | le mal que port. |

Example 14. annotated score of Machaut's V27/24, "En ma cuer;" cantus only, metrically interpreted at the *modus* level, directed progressions circled

As mentioned above, the only directed progression that falls on the weak beat 2 is a contrapuntally weak, inverted one ($m3 \rightarrow P1$) in m. 21, which provides an anticipatory taste of the final F. The F of m. 21 is contrapuntally, metrically, lyrically, and formally subordinate to the $M6 \rightarrow P8$ directed progression that closes the A section on beat 1 of the following measure.

It is only conventional that each section of this song, A and B, should close on beat 1. Cadencing on a strong beat prepares both of this *chanson baladées'* formal junctures to begin with an upbeat, further reason to prefer the anacrusic opening of the dance shown above.

The consistent placement of preparatory gestures can provide a musical cue for metric interpretation, not only at the beginning but as found throughout a danced song. In V27/24, the same gesture in m. 1, identified above for its melodic and rhythmic directedness, returns

often across the song, as at mm. 5, 11, 15, 17, 27, and in varied form at 15 and 23. That in every instance this melodic preparation falls on a weak beat confirms the subordinate role of that beat in the emergence of duple meter and argues strongly for beginning this virelai with an anacrusis. In V31/28, the opening figure of m. 1 does not appear so melodically driven as the ascending leading tone of Virelais 27/24 and 32/29. Rather, its function becomes clear through Machaut's consistent usage of it as cadential preparation. The formal cadences that close both the A and B sections (mm. 21-22 and 34-35) are motivic repetitions of m. 1, as summarized in **Example 15** (beat labels 2 and 1 apply vertically for all staves). Insofar as cadential arrivals are conventionally marked by a strong beat, their preparatory gesture should be interpreted as weak, and this norm can be backdated to establish the beginning of the song.

beats

2 1

1

8 Plus du - re

21

8 a - mi - tie.

34

8 sous-ri - ant.

Example 15. weak beat prep to contrapuntal arrival in Machaut's V31/28, "Plus dure que un dyamant," mm. 1-2, 21-22 (end of A), and 34-5 (end of B); cantus only

The final danced song in triple time, "Tres bonne et belle" (V26/23), is irregular at the *modus* level regardless of how it is begun. The cantus melody opens with the descending pentachord G-C, decorates that C (^1) with both neighbors (D in m. 2 and B in m. 3), then descends again towards G (mm. 4-5) to start the next phrase. A compressed variation of this melodic framework returns within the A section (mm. 16-19), reflecting the poetry's rhyme scheme (-*ueil*). But this melodic return comes against the prevailing metrical (*modus*) entrainment. Not only is the melody now metrically unstable—strong and weak beats are reversed—but it is also recontextualized by a different accompaniment in the two lower voices.

Example 16 presents these two passages in question for comparison, with aligned beats at the *modus* level. The metric disturbance here may shed light upon Machaut's courtly love lyric.

While the lover's eyes are smitten with the beloved in the first phrase (*Tres bonne et belle, mi oueil joyeuse pasture prennent en vostre figure*), the heart of the second phrase will be stung by the pains of unrequited love rather than enjoy any sweet nurturing (*et mes cuers en vostre accueil vie et douce norreture*).

1 1 2 1 2 1

8 Tres bonne et bel - le, mi oueil | Joy-

CONTRATENOR

TENOR

6 3 — 8 5

15 1 2 1 2 1

8 Et mes cuers en vostre ac - cueil |

6 3 — 8 5

Example 16. melody recontextualized in Machaut's V26/23, "Tres bonne et belle;" mm. 1-5 and

15-19, metrically interpreted at the *modus* level and directed progressions marked

The uneasy metrical setting I am suggesting rests upon the norm of imperfect *modus* groupings (*tempus* being what it may). The directed intervals harmonizing the preparatory word "et" that begins the second phrase (imperfect sixth and third in m. 15) suggests a weak beat rather than a strong one. If a strong beat were to come down on m. 16, the perfect resolution of that charged harmony would kick off this phrase more parallel to the first. If this

step beat would be to advantage, the analyst may be tempted to interpret a *modus* perfection of measures 13-15, a triple group whose final measure is the preparatory 15 and which all the more forcefully begins the second phrase at m. 16 on a proper downbeat in all parameters. However, mm. 13-14 form a clear cadence (E6/3 to D8/5) which ought to proceed weak—strong by the same logic. This in turn pushes back an initiating downbeat to m. 12 and the *modus* perfection somewhere between mm. 9-12, an exercise left to the reader. In any event, the entire refrain cannot repeat on the same metrical footing when it returns at the end of the virelai form because it is an odd number of measures. If the final cadence (m. 23) is to be a strong beat, respecting this as the very strongest metrical convention in the style, m. 1 really ought to be a weak beat and something in the middle will have to be perfected.

Some speculation about dancing Machaut's "danced songs"

Social dancing was an integral part of both courtly and country life in medieval France and beyond.⁴⁵ Indeed, "dance was ever-present in the lives of western Europeans from the late

⁴⁵ For a rich and broad sweep of dance's importance to western European society—artistic, political, and intellectual—from 1250-1750, see Nevile 2008a and 2008b. Renberg 2022 recounts the fall of dance in early medieval texts from worshipful to sinful (as gendered bodies in motion), while Dickason 2021 explores the reclamation of dance from transgression to integral part of medieval religious practice. Chaganti 2018 details how the central position of dance in late medieval society informed the creation and reception of lyric poetry. Page 1989, esp. Ch. 5, views the French *carole*'s social context from the perspective of an evolving clergy. Straple-Sovers 2023 is instructive on the challenges and authenticity of recreating medieval dance practice, given the dearth of sources; see also 2021 on reading bodily movement—with caroling as cautionary example—in medieval

Medieval period to the middle of the eighteenth century” (Nevile 2008a, 2). The *carole*—after the Old French verb *caroler*, to dance—was the simplest and most common type of dance, well established in France by the mid-twelfth century (Mullally 1986, 224).⁴⁶ Medieval French writings on a wide range of topics mention the *carole* and illuminated romances provide numerous depictions of it (such as **Example 2**). Alas, no European dance manuals exist before the fifteenth century (Nevile 2008a, 7, and Dickason 2021, 4).⁴⁷ Securely matching fourteenth-century music to dance steps may ever remain elusive, and it is not attempted here.⁴⁸ I

literature. Arcangeli 1994 reviews the colorful denunciation of dance by medieval clergy and moralists, not limited to claims of adultery, gluttony, paganism, lechery, and disregard for the Sabbath.

⁴⁶ Also *karole* and *querole*. Latin: *chorea*. The term was used both then and now to refer to the dance itself and to the music that accompanied dancing in a *carole*. McGee 2012, 399, notes that “scholars in all fields use the term [carole] with a vague and sometimes contradictory understanding of exactly to what it refers.” Stevens 1986, 175, calls the *carole* “a dance-idea waiting to be realized in various forms.” Dickason 2021, 5, points to the array of activities inseparably connoted by Latin terms for dancing, including not only bodily movement but singing and exalting God. Mullally 2011 discusses the etymology, choreography, and social participants of this simple dance without detailed analysis of musical structure. In a highly critical review, McGee 2012 casts doubt on Mullally’s dismissal of commonly accepted views of caroling, his biased selection and weighing of evidence on the dance and its accompanying music, and his specific application of the hypothetical dance steps he posits for the *carole*.

⁴⁷ Ward 1976 and Dean Smith 1937 provide accounts of a fifteenth-century handbook on the *basse danse* that introduces step patterns. Little 1975 discusses a Baroque dance manual more closely matched to the given music, including implications for analysis and musical performance.

⁴⁸ Rimmer 1985 is a speculative attempt at matching specific dance step patterns with medieval *trouvère* repertory. Her choreography rests upon the dubious assumptions that the steps of fifteenth-century *branles* can be backdated to “chain and round dances [*caroles*] of which we know practically nothing” yet which are presumed to have “identical step elements” (26).

endeavor only to begin articulating some considerations that might arise if coordinating rhythmic movement (whatever it may be) with singing the genre Machaut dubbed “danced songs.” By engaging in such speculation, I do not claim that these works were danced in Machaut’s time. Though dance cannot be seen in manuscript or modern score, nor in concerted performance today, the interplay of movement and song may have been top of mind to the medieval composer, singer, and dancer. It seems only natural that a dance and its accompanying music would evolve in response to each other in subtle ways that we lack evidence to recreate authentically.

I will seize upon only one particular element of a considerable portion of Machaut’s “danced songs” that is suggestive of, if not a relic of, dance awareness. According to Aristotelian doctrine, “time was only rendered sensible as a description or quantification of enacted motions or changes” (Grant 2014, 15). Recall of Virelai 14, **Example 5** above, that the first musical event is a dotted quarter rest, a transcription of the semibreve rest in all sources (Mach A, C, G, and Vg); omitted by Ludwig 1926 and restored by Schrade 1956.⁴⁹ The first sung pitch occurs on the weak beat after rest.⁵⁰ **Example 17** shows this song’s opening in the clear notation of Mach C.

⁴⁹ V14 joins V4, 27/24, and 36/30 as a small but not inconsiderable set of virelais that begin with a semibreve beat of rest in all manuscripts. Ludwig reproduces only that of V27/24, likely to counterpoint the tenor’s entrance, while Schrade begins both V27/24 and 36/30 with a beat of rest (both polyphonic) but not V4 (monophonic).

⁵⁰ Little 1975, 114 and 119, discusses a French Baroque dance manual, one of the first to inlay music notation with dance steps, in which pickup notes are danced with an introductory plié before the steps proper begin.



Example 17. a breve and semibreve rest perhaps initiate dancing before the singing of

Machaut's V14; excerpt of *F-Pn* fr. 1586, f. 153v

While an opening rest may make little difference to singing alone, its semibreve duration joins with the first note to complete the characteristic mensural unit of imperfect breves in Machaut's "danced songs." More importantly, the opening rest establishes an entrainment by which the first sung pitch is felt as metrically weak in a notation lacking bar lines or other explicit metrical reference. I further speculate that a rest is also suggestive of reserving metered space for dancers to begin motion before a pickup note is sung on the offbeat.⁵¹ Even more remarkable is the breve rest preceding the semibreve rest just discussed, the breve representing a full measure in transcription (noted but omitted by Ludwig and by my **Example 5** because it did not affect beat counts at the *tempus* level). Because both these opening rests appear consistently in all sources, the empty breve (measure) would appear to serve an important function in the *modus* organization of this song. If m. 1 of my transcription

⁵¹ Saltzstein 2010, 246-50, summarizes then challenges the "scholarly genealogy" of 13th-century notated (intertextual) refrains as recording an otherwise lost oral tradition of dance songs (*caroles*), locating them instead within a practice of literate quotation and citation, one that elevated French as a literary language.

is taken as a hypermetrical upbeat, adopting the empty breve as the downbeat of m. 0 (not shown), then all formal sections of V14 cadence more emphatically on *modus* downbeats (mm. 14, 18, and 20 of Ludwig). If, on the other hand, m. 1 is felt to start the song on a *modus* downbeat, all formal boundaries—indeed, the end of the song—close on a weak beat at that level.

Four of Machaut's thirty-three *chansons baladées* in Ludwig (V5, 9, 28/25, and 36/30) are sung beginning with a single eighth note pickup, and this subset displays a similar runup with rests. Lacking a full breve of preparation like in V14, these singleton minims are perhaps brief enough not to confuse mensural reckoning and, relatedly, to be sung before dancing commences on the ensuing downbeat. In all manuscripts, V5, 9, and 28/25 begin with one or two minim rests (in imperfect and perfect time, respectively), enough to complete a full semibreve beat together with their single minim pickup note (excepting Mach E's notation of V28/25). V36/30 begins with one semibreve rest and then a minim rest in all sources (Mach A, G, and Vg); that is, a full beat of rest followed by enough rest to complete a second beat of imperfect time with the single pickup note, a duration of silence approaching that of V14.⁵²

Intriguingly, beginning on a weak beat is a trait peculiar to the *chansons baladées* among Machaut's entries in the *formes fixes*. There are no examples amongst the ballades or rondeaux. To include his other song genres—the *lais*, a *complainte*, and a *chanson royale*—adds potentially only one other song that begins with a pickup. Ludwig interprets Lais 1, “Loyaute, que point ne delay,” to begin on beat 2 of 6/4 without the guide of a beat 1 rest in

⁵² Ludwig represents none of these rests while Schrade restores them all.

any manuscript source. Because Schrade sets the same song in 3/4, bringing Machaut's older *modus* notation into line with the *tempus* beats of other genres, the question is moot.⁵³ The prospect of dance may suggest why measured silences, a special feature of about 10% of Machaut's danced songs, were worth the trouble of notating for a medieval composer and his audience. Opening rests were crucial not only for mensural reckoning but also, perhaps, for dancing insofar as they created space for movement on those occasions when *chanson* and *baladée* commenced separately. Yet because this subset of Machaut's virelais includes polyphonic examples, a complexity that likely excluded social dancing, offbeat pickups may have figured as a generic artefact of dance music at this point.

Even if Machaut's *chansons baladées* overstepped their practical dance origins, Seeta Chaganti makes the case that social dancing so permeated medieval French society that lyrics like Machaut's were both crafted and read through the virtual embodiment of dance (Chaganti 2018, 2-4 and passim).⁵⁴ The virtual embodiment of dance is the sense in which poets took on the norms and limitations of dance music as a scaffolding for the creation of new lyrics, and it is not a far leap to suppose that the genre Machaut proclaimed the "danced song" could encode features of dance music known to him. Judith Peraino similarly claims that whether a lyric was

⁵³ Concerning mensural notational options in the *Ars nova*, Jacques de Liège (1320s) observes that "the perfect breve has the same value as had the perfect long for the Ancients," with consequences for modern transcription; see Maw 2004, 47.

⁵⁴ When Chaganti 2018, 17, notes that "Machaut and Deschamps wrote lyrics that were mindful of the aesthetics of dance and dance accompaniment," this does not mean that these lyrics were necessarily danced but rather that they embrace certain markers of that tradition, analogous to a bourree by Bach or, still further removed, the "ballades" of Chopin.

actually sung or just read “has less to do with a change in the formal properties of the poetry than with a change in concept, namely, that a lyrical poem with a refrain and an origin in public dance songs could be a complete expression as read—without musical performance” (Peraino 2011, 259). Whether a given work was destined to be danced actually or not, its metrical grouping structure may bear the generic footprint of dance.

The metrical flexibility explored in this study, as I have noted, is taken as evidence of a highly developed style rooted in dance. *Ars nova* notation’s potential for ongoing grouping changes raises questions of whether and how dancers of the time might have interpreted and embodied flexible groupings and metrical inconsistencies, to which we have no sure medieval answers. The primacy of binary grouping in Machaut’s *chansons baladées*, taken as a stylistic norm, prompts special interpretation of triple groupings. In this study I have extended this simple principle to draw attention to any change from an entrained grouping pattern at any level of metrical grouping. I then proposed connections between passages of metrical ambiguity, marked off by such change, and strategic tonal articulation. A similar off-kilter status was proposed for large-scale formal and small-scale motivic repeats set in opposing metric positions as a given song unfolds. The subtle complexity of these issues and the presence of polyphonic accompaniment in a number of Machaut’s *chansons baladées* likely witness a shift from physical to virtual embodiment of dance in these songs. While thus enlarging the possibilities of the danced song as a genre, the composer clung to nomenclature that ensured dance could not be out of mind for performers and auditors of his era and for scholars today. Further research on the interconnections between dance, music, and medieval meter can guide vital interpretation of these works in our own time.

MACHAUT SOURCES

A: Paris, Bibliothèque Nationale de France, fonds français 1584

B: Paris, Bibliothèque Nationale de France, fonds français 1585

C: Paris, Bibliothèque Nationale de France, fonds français 1586

E: Paris, Bibliothèque Nationale de France, fonds français 9221

F-G: Paris, Bibliothèque Nationale de France, fonds français 22545-6

Vg: Ferrell-Vogüé, private ownership of James E. and Elizabeth J. Ferrell

WORKS CITED

Arcangeli, Alessandro. 1994. "Dance under Trial: The Moral Debate 1200-1600." *Dance Research: The Journal of the Society for Dance Research* 12 (2): 127-155.

Alexander, Jonathan J. G. 1996. "Dancing in the Streets." *The Journal of the Walters Art Gallery* 54: 147-162.

van den Boogaard, Nicolaas Hendricus Johannes. 1969. *Rondeaux et refrains du xiie siècle au début du xive: collationnement, introduction, et notes*. Paris: Éditions Klincksieck.

Boone, Graeme M. 2000. "Marking Mensural Time." *Music Theory Spectrum* 22: 1-43.

Boulton, Maureen Barry McCann. 1993. *The Song in the Story: Lyric Insertions in French Narrative Fiction, 1200-1400*. Philadelphia: University of Pennsylvania Press.

- Butterfield, Ardis. 2002. *Poetry and Music in Medieval France: from Jean Renart to Guillaume de Machaut*. Cambridge: Cambridge University Press.
- Chaganti, Seeta. 2018. *Strange Footing: Poetic Form and Dance in the Late Middle Ages*. Chicago: University of Chicago Press.
- Coldwell, Maria. 1981. "Guillaume de Dole and Medieval Romances With Musical Interpolations." *Musica Disciplina* 35: 55-86.
- Dean Smith, Margaret. 1937. "A Fifteenth Century Dancing Book." *Journal of the English Folk Dance and Song Society* 3 (2): 100-109.
- Desmond, Karen. 2018. *Music and the modern, 1300-1350*. Cambridge: Cambridge University Press.
- Dickason, Kathryn. 2021. *Ringleaders of Redemption: How Medieval Dance Became Sacred*. New York: Oxford University Press.
- Dillon, Emma. 2015. "Unwriting Medieval Song." *New Literary History* 46 (4): 595-622.
- Earp, Lawrence M. 2012. "Declamation as Expression in Machaut's Music." In *A Companion to Guillaume de Machaut*, ed. Deborah McGrady and Jennifer Bain, 209-238. Boston: Brill.
- _____. 2005. "Declamatory Dissonance in Machaut." In *Citation and Authority in Medieval and Renaissance Musical Culture*, ed. Suzannah Clark and Elizabeth Eva Leach, 102-122. Rochester: The Boydell Press.
- _____. 1995. *Guillaume de Machaut: A Guide to Research*. New York: Garland Publishing.
- _____. 1991. "Lyrics for Reading and Lyrics for Singing in Late Medieval France: The Development of the Dance Lyric from Adam de la Halle to Guillaume de Machaut." In

- The Union of Words and Music in Medieval Poetry*, ed. Rebecca A. Baltzer, Thomas Cable, and James I. Wimsatt, 101-31. Austin: University of Texas Press.
- Eustace, Frances and Pamela King. 2011. "Dances of the Living and the Dead: A Study of *Danse Macabre* Imagery within the Context of Late-Medieval Dance Culture." In *Mixed Metaphors: The Danse Macabre in Medieval and Early Modern Europe*, ed. Sophie Oosterwijk and Stefanie Knöll, 43-72. Newcastle upon Tyne: Cambridge Scholars Publishing.
- Fuller, Sarah. 1992. "Tendencies and Resolutions: The Directed Progression in 'Ars Nova' Music." *Journal of Music Theory* 36 (2): 229-258.
- _____. 1981. "Theoretical Foundations of Early Organum Theory." *Acta Musicologica* 53 (1): 52-84.
- Grant, Roger Matthew. 2014. *Beating Time and Measuring Music in the Early Modern Era*. Oxford: Oxford University Press.
- Hoppin, Richard H. 1960. "Notational Licences of Guillaume de Machaut." *Musica Disciplina* 14: 13-27.
- Huot, Sylvia. 1987. *From Song to Book: The Poetics of Writing in Old French Lyric and Lyrical Narrative Poetry*. Ithaca: Cornell University Press.
- Kohn, Karl. 1981. "The Renotation of Polyphonic Music." *The Musical Quarterly* 67 (1): 29-49.
- Lavacek, Justin. 2022. "Hidden Colouration: Deep Metrical Flexibility in Machaut." *Plainsong and Medieval Music* 31/2: 143-167.

- Leach, Elizabeth Eva. 2011. *Guillaume de Machaut: Secretary, Poet, Musician*. Cornell: Cornell University Press.
- _____. 2000. "Counterpoint and Analysis in Fourteenth-Century Song." *Journal of Music Theory* 44 (1): 45-79.
- Leech-Wilkinson, Daniel. 1991. "Not Just a Pretty Tune: Structuring Devices in Four Machaut Virelais." *Sonus* 12 (1): 16-31.
- Leguy, Sylvette. 1977. *Oeuvres Complètes*: Vol. 1. Paris: Le Droict Chemin de Musique.
- Leo, Domenic. 2005. "Authorial Presence in the Illuminated Machaut Manuscripts." Ph.D. diss., New York University.
- Lerdahl, Fred and Ray Jackendoff. 1983. *A Generative Theory of Tonal Music*. Cambridge, MA: MIT Press.
- Little, Meredith Ellis. 1975. "The Contribution of Dance Steps to Musical Analysis and Performance: 'La Bourgogne.'" *Journal of the American Musicological Society* 28 (1): 114.
- Ludwig, Friedrich, ed. 1926. *Musikalische Werke Vol. 1: Balladen, Rondeaux, und Virelais*. Leipzig: Breitkopf & Härtel.
- Maw, David. 2013. "'Bona Cadentia Dictaminum:' Reconstructing Word Setting in Machaut's Songs." *Music & Letters* 94/3: 383-432.
- _____. 2004. "'Trespasser mesure:' Meter in Machaut's Polyphonic Songs." *The Journal of Musicology* 21 (1): 46-126.

- _____. 2002. "Meter and Word Setting: Revising Machaut's Monophonic Virelais." *Current Musicology* 74: 69-102.
- McGee, Timothy J. 2012. Review of *The Carole: A Study of a Medieval Dance* by Robert Mullally. *Music & Letters* 93 (3): 399-401.
- _____. 1989. "Medieval Dances: Matching the Repertory with Grocheio's Descriptions." *The Journal of Musicology* 7 (4): 498-517.
- Moll, Kevin N. 1998. "Voice Function, Sonority, and Contrapuntal Procedure in Late Medieval Polyphony." *Current Musicology* 64: 26-72.
- Morrison, Jr., Clint. 2021. "Carole Dancing in *Sir Gawain and the Green Knight*." *Arthuriana* 31 (3): 21-39.
- Mullally, Robert. 2011. *The Carole: A Study of a Medieval Dance*. New York: Routledge.
- _____. 1998. "Johannes de Grocheo's 'Musica Vulgaris.'" *Music & Letters* 79 (1): 1-26.
- _____. 1986. "Cançon de Carole." *Acta Musicologica* 58 (2): 224-231.
- Nevile, Jennifer. 2008a. "Dance in Europe 1250-1750." In *Dance, Spectacle, and the Body Politick, 1250-1750*, ed. Jennifer Nevile, 7-64. Bloomington: Indiana University Press.
- _____. 2008b. "Dance Performance in the Late Middle Ages: A Contested Space." In *Visualizing Medieval Performance: Perspectives, Histories, Contexts*, ed. Elina Gertsman, 295-310. Burlington: Ashgate.
- Page, Christopher. 2000. "Around the Performance of a 13th-Century Motet." *Early Music* 28 (3): 343-348 + 351-357.

- _____. 1997. *Discarding Images: Reflections on Music & Culture in Medieval France*. Oxford: Clarendon Press.
- _____. 1989. *The Owl and the Nightingale: Musical Life and Ideas in France 1100-1300*. Berkeley: University of California Press.
- Peraino, Judith A. 2011. *Giving Voice to Love: Song and Self-Expression from the Troubadours to Guillaume de Machaut*. Oxford: Oxford University Press.
- Renberg, Lynneth Miller. 2022. *Women, Dance and Parish Religion in England, 1300-1640: Negotiating the Steps of Faith*. Woodbridge: Boydell Press.
- Rimmer, Joan. 1991. "Medieval Instrumental Dance Music." *Music & Letters* 72 (1): 61-68.
- _____. 1985. "Dance Elements in Trouvère Repertory." *Dance Research: The Journal of the Society for Dance Research* 3 (2): 23-34.
- Robbins, Harry W., trans. 1962. *The Romance of the Rose*. New York: E. P. Dutton.
- Saltzstein, Jennifer. 2010. "Relocating the Thirteenth-Century Refrain: Intertextuality, Authority and Origins." *Journal of the Royal Musical Association* 135 (2): 245-79.
- Schrade, Leo. 1956. *The Works of Guillaume de Machaut: Part 2*. Monaco: L'Oiseau-Lyre.
- Stevens, John. 1986. *Words and Music in the Middle Ages*. Cambridge: Cambridge University Press.
- Stokes, Jordan. 2014. "In Search of Machaut's Poietics: Music and Rhetoric in Le Remede de Fortune." *The Journal of Musicology* 31/4: 395-430.
- Straple-Sovers, Rebecca. 2023. "'A Feather on the breath of God: Medieval legacies in modern choreography." *Postmedieval* 14: 593-619.

- _____. 2021. "Kinesic Analysis: A Theoretical Approach to Reading Bodily Movement in Literature." In *The Cursed Carolers in Context*, ed. Lynne Miller Renberg and Bradley Phillis, 21-38. London: Routledge.
- Strunk, Oliver, ed. 1997. "Marchetto's Pomerium." In *Source Readings in Music History: The Early Christian Period and the Latin Middle Ages*, 160-171. New York: Norton.
- Ward, John M. 1976. "The Maner of Dauncying." *Early Music* 4 (2): 127-140+142.